

# MOSAIC: Model-Aware Simultaneous Anthropometry, Inertia and Control

## Estimation From Repeated Motion Captures

Methods Reference for Forward-Dynamics Matching in UpstreamDrift

UpstreamDrift Estimation Working Reference

Revision 2026-10-05 (separate research and experiment record; see §1)

### Abstract

Matching a forward-dynamics model to measured kinematics requires three things to be determined at once: the subject’s segment geometry and inertial parameters, the joint inputs that drove each trial, and a kinematic trajectory consistent with both the observations and the equations of motion. This reference specifies MOSAIC, an estimator that exploits two exact structural facts of rigid-body dynamics—the control-affine form  $\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$  and the linearity of inverse dynamics in the inertial parameters,  $\mathbf{Y}(\mathbf{q}, \mathbf{v}, \mathbf{a})\boldsymbol{\pi} = \mathbf{B}\mathbf{u} + \mathbf{J}_c^\top \boldsymbol{\lambda}$ —to turn the joint problem into a variable-projection Gauss–Newton method over a physically consistent inertial manifold. Inputs, cross-trial torque templates and contact wrenches are eliminated exactly by a sparse linear inner solve; kinematics, geometry and inertia are updated by a damped reduced Gauss–Newton step whose dynamics Jacobians are exact in the inertial direction. The method clarifies what kinematics can and cannot identify (base-parameter combinations, a global mass gauge, the massless-body degeneracy of minimum-effort fitting), states where the missing information comes from in golf (the club as a known-inertia force sensor, the unactuated base rows, ground reactions), and places the zero-torque and zero-velocity counterfactuals (ZTCF/ZVCF) in the same algebra as exact row and column projections of the regressor. A NumPy reference implementation on an analytic planar chain recovers geometry to 0.1 mm, observable inertial parameters to 1–3 %, torques to 20 % RMS, and passes an open-loop forward-replay gate (0.03 rad RMS over 0.75 s) in 37 outer iterations and about six seconds for two trials. No human capture has been qualified with the method yet; the qualification gates and the engine integration path are specified so that claim can be earned, not asserted.

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## 1 Status, Scope and Governance

This document is an editable standalone research and experiment record in the sense of `AGENTS.md` (“Modeling Reference Documentation”); it is not the engineering design manual, whose only editable source is `manuals/upstreamdrift`. It is the calculation-level reference for the MOSAIC estimator implemented in `src/shared/python/estimation/mosaic/` and for the review of the Dynamics-Informed Motion Estimation epic (DIME, #11421) recorded in §12. It complements, and is cross-linked from, the GS3DX record `docs/research/simscape_matching_reference/`.

### Qualification status (truthful).

- Verified: all equations of §3–§7 against closed-form double-pendulum dynamics and synthetic ground truth (36 behavioural tests, `tests/unit/estimation/mosaic/`).
- Measured: the planar results of §9 on the analytic fixture (capture-A/capture-O style noise levels, no engine).
- Not yet done: any fit of a human capture (capture-A, capture-O), any 3-D floating-base model, any contact-wrench estimation, any engine-backed regressor. These are the child issues of the follow-up epic (§13). Nothing here qualifies the GS3DX IK videos.

## 2 Problem Statement and Notation

A subject performs  $K$  trials (swings). Trial  $k$  is observed at times  $t_1^k < \dots < t_{T_k}^k$  through markers or 2-D keypoints  $\mathbf{y}_t^k$  with covariance  $\mathbf{R}_t^k$ , optionally with ground-reaction wrenches. A rigid multibody model with configuration  $\mathbf{q} \in \mathcal{Q}$  (a Lie group when a floating base or spherical joints are present), velocity  $\mathbf{v} \in \mathbb{R}^{n_v}$  and acceleration  $\mathbf{a}$  obeys

$$\mathbf{M}(\mathbf{q}; \boldsymbol{\pi}, s) \mathbf{a} + \mathbf{h}(\mathbf{q}, \mathbf{v}; \boldsymbol{\pi}, s) = \mathbf{B}\mathbf{u} + \mathbf{J}_c(\mathbf{q}; s)^\top \boldsymbol{\lambda}, \quad (1)$$

with  $\mathbf{h} = \mathbf{C}\mathbf{v} + \mathbf{g}$ , actuation map  $\mathbf{B} \in \mathbb{R}^{n_v \times n_u}$  ( $n_u < n_v$ : the floating base and any passive joints are unactuated), contact Jacobian  $\mathbf{J}_c$  and wrenches  $\boldsymbol{\lambda}$ . Unknowns:

| Symbol                                       | Shared/per-trial | Meaning                                                                                                   |
|----------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------|
| $s \in \mathbb{R}^{n_s}$                     | shared           | geometry: segment lengths, marker attachment offsets, joint-axis offsets                                  |
| $\boldsymbol{\pi} \in \mathbb{R}^{10n_b}$    | shared           | inertial parameters ( $m_i, \mathbf{h}_i = m_i \mathbf{c}_i, \mathbf{I}_i$ ) about each body-frame origin |
| $\boldsymbol{\theta}$                        | shared           | unconstrained coordinates of $\boldsymbol{\pi}$ on the consistency manifold (§5)                          |
| $\mathbf{C}^k \in \mathbb{R}^{C \times n_v}$ | per trial        | spline coefficients of the kinematic trajectory $\mathbf{q}^k(t) = \mathbf{P}(t)\mathbf{C}^k$             |
| $\mathbf{u}_t^k$                             | per trial        | net joint inputs (torques; activations via a declared actuator model)                                     |
| $\bar{\mathbf{u}}(\phi)$                     | shared           | phase-indexed torque template coupling trials                                                             |
| $\boldsymbol{\lambda}_t^k$                   | per trial        | contact wrenches (friction-cone constrained)                                                              |

$\boldsymbol{\xi} = (\mathbf{C}^1, \dots, \mathbf{C}^K, s, \boldsymbol{\theta})$  are the *outer* variables,  $w = (\mathbf{u}, \bar{\mathbf{u}}, \boldsymbol{\lambda})$  the *inner* variables. Units are SI throughout; residuals are whitened before any norm is taken so that radians and metres are never mixed (cf. ch. 21 of the AffineDrift companion).

### 3 Structure Exploited

#### 3.1 Control-affine form and the inertial regressor

Equation (1) is control-affine:  $\dot{\mathbf{x}} = f(\mathbf{x}) + G(\mathbf{x})\mathbf{u}$  with  $f = (\mathbf{v}, -\mathbf{M}^{-1}\mathbf{h} + \mathbf{M}^{-1}\mathbf{J}_c^\top \boldsymbol{\lambda})$  and  $G = (0, \mathbf{M}^{-1}\mathbf{B})$ . Independently, the left-hand side of (1) is *linear* in  $\boldsymbol{\pi}$  (Atkeson–An–Hollerbach; Gautier–Khalil):

$$\mathbf{M}(\mathbf{q})\mathbf{a} + \mathbf{h}(\mathbf{q}, \mathbf{v}) = \mathbf{Y}(\mathbf{q}, \mathbf{v}, \mathbf{a}; s) \boldsymbol{\pi}, \quad \mathbf{Y} \in \mathbb{R}^{n_v \times 10n_b}. \quad (2)$$

Both facts hold for every rigid multibody engine (Pinocchio exposes  $\mathbf{Y}$  as `computeJointTorqueRegressor`; MuJoCo/MJX and Drake give it by differentiating inverse dynamics with respect to the body inertias). MOSAIC consumes only  $\mathbf{Y}$ ,  $\mathbf{B}$ ,  $\mathbf{J}_c$  and marker forward kinematics, so it is engine-agnostic by construction.

#### 3.2 ZTCF and ZVCF as exact projections of the regressor

Because  $\mathbf{Y}$  is affine in  $\mathbf{a}$  and  $\mathbf{h} = \mathbf{Y}(\mathbf{q}, \mathbf{v}, 0)\boldsymbol{\pi}$ , the drift and its parts are algebraic objects of the same regressor:

$$\mathbf{a}_{\text{ZTCF}}(t) = -\mathbf{M}(\mathbf{q})^{-1} \mathbf{Y}(\mathbf{q}, \mathbf{v}, 0) \boldsymbol{\pi} + \mathbf{M}^{-1} \mathbf{J}_c^\top \boldsymbol{\lambda}_0, \quad (3)$$

$$\mathbf{a}_{\text{ZVCF}}(t) = -\mathbf{M}(\mathbf{q})^{-1} \mathbf{Y}(\mathbf{q}, 0, 0) \boldsymbol{\pi} + \mathbf{M}^{-1} \mathbf{J}_c^\top \boldsymbol{\lambda}_{00}, \quad (4)$$

$$\mathbf{a}_{\text{ZTCF}} - \mathbf{a}_{\text{ZVCF}} = -\mathbf{M}^{-1} [\mathbf{Y}(\mathbf{q}, \mathbf{v}, 0) - \mathbf{Y}(\mathbf{q}, 0, 0)] \boldsymbol{\pi} \quad (\text{velocity-dependent drift}). \quad (5)$$

Here  $\boldsymbol{\lambda}_0$  ( $\boldsymbol{\lambda}_{00}$ ) are the contact reactions recomputed under zero input (zero input and zero velocity) from the same constraint equations; in free flight they vanish. The measured acceleration decomposes exactly as  $\mathbf{a}_{\text{meas}} = \mathbf{a}_{\text{ZTCF}} + \mathbf{M}^{-1}\mathbf{B}\mathbf{u} + \mathbf{M}^{-1}\mathbf{J}_c^\top(\boldsymbol{\lambda} - \boldsymbol{\lambda}_0)$ , which is the AffineDrift pointwise decomposition  $\dot{\mathbf{x}} - f(\mathbf{x}) = G(\mathbf{x})\mathbf{u}$  with contact made explicit.

**Canonical nomenclature (fixing the inconsistencies found in the repository, §12).**

- **ZTCF**:  $\mathbf{u} = 0$ , full state kept, constraints kept, reactions recomputed. *Pointwise* unless the word “trajectory” is attached; the integrated object  $\mathbf{x}(t_0) + \int f(\mathbf{x}_{\text{ZTCF}}(\tau))d\tau$  is a different curve (it follows its own flow, not the measured state) and must be called the *ZTCF trajectory*.

- **ZVCF**:  $\mathbf{v} = 0$  and  $\mathbf{u} = 0$  at each instant (configuration-only drift: gravity plus constraint reactions). The variant that keeps the control is *zero-velocity, control-preserved acceleration* and is never abbreviated ZVCF.
- Passive impedance declared in the model (joint damping, ligament stiffness) is part of  $f$ ; a feedback servo is part of  $\mathbf{u}$  and is removed by the ZTCF.

### 3.3 Where the information lives: unactuated rows

Let  $\mathbf{S}_\perp$  select the unactuated coordinates ( $\mathbf{S}_\perp \mathbf{B} = 0$ ). Then (1) gives, with *no unknown input*,

$$\mathbf{S}_\perp \mathbf{Y}(\mathbf{q}, \mathbf{v}, \mathbf{a}) \boldsymbol{\pi} = \mathbf{S}_\perp \mathbf{J}_c^\top \boldsymbol{\lambda}. \quad (6)$$

These six-per-node (floating base) or one-per-node (passive pivot) equations are the entire torque-free information kinematics carries about  $\boldsymbol{\pi}$  (Ayusawa–Venture–Nakamura). Equivalently, in ZTCF language,  $\mathbf{S}_\perp \mathbf{M}(\mathbf{a}_{\text{meas}} - \mathbf{a}_{\text{ZTCF}}) = \mathbf{S}_\perp \mathbf{J}_c^\top (\boldsymbol{\lambda} - \boldsymbol{\lambda}_0)$ : the “torque-independent feasibility criterion” of DIME-16 is the unactuated projection of the regressor equation, not an additional likelihood. The actuated rows determine  $\mathbf{u}$  once  $\boldsymbol{\pi}$  is known; they carry information about  $\boldsymbol{\pi}$  only through priors on  $\mathbf{u}$ .

### 3.4 The club as a known-inertia force sensor

In golf the distal body is a manufactured object whose inertial parameters are measurable to better than 1 % off-line. With  $\boldsymbol{\pi}_{\text{club}}$  known and its kinematics observed, the club’s Newton–Euler equations return the hand wrench exactly, and the unactuated rows (6) become *inhomogeneous* in the unknown human parameters,  $\mathbf{S}_\perp \mathbf{Y}_{\text{human}} \boldsymbol{\pi}_{\text{human}} = -\mathbf{S}_\perp \mathbf{Y}_{\text{club}} \boldsymbol{\pi}_{\text{club}} + \dots$ , which removes the global scale gauge without a force plate. This is the golf-specific anchor used in §9; for a humanoid the analogue is a known payload or the measured ground wrench.

## 4 Identifiability

**Proposition 1** (Scale gauge). *If no external wrench is measured and no anchor is imposed, the map  $(\boldsymbol{\pi}, \mathbf{u}, \boldsymbol{\lambda}) \mapsto (\alpha \boldsymbol{\pi}, \alpha \mathbf{u}, \alpha \boldsymbol{\lambda})$  leaves every residual of (1) invariant for all  $\alpha > 0$ . Kinematics alone cannot identify absolute mass.*

**Proposition 2** (Base parameters). *Only the row space of the stacked matrix  $[\mathbf{S}_\perp \mathbf{Y}(\mathbf{q}_t, \mathbf{v}_t, \mathbf{a}_t)]_t$  is observable from the unactuated rows. A distal link’s mass lumps into the proximal link’s first moment and inertia ( $h_{0x} + m_1 l_0$ ,  $I_0 + m_1 l_0^2$  in the planar chain); the orthogonal complement is determined by the prior only. The reference implementation reports this subspace (*structural\_observability*) and the Fisher information after eliminating inputs (*data\_information*).*

**Proposition 3** (Massless-body degeneracy of minimum-effort fitting). *With free  $\boldsymbol{\pi}$  constrained only by positivity, a free kinematic trajectory and any prior that penalises input magnitude, the configuration “all mass at the unactuated pivot, every other body massless,  $\mathbf{u} \equiv 0$ ” attains zero dynamics residual and zero effort for every trajectory. Without an anchor or a credible anthropometric prior the estimator converges to it.*

*Sketch.* A body whose mass sits at the origin of the unactuated joint contributes  $\mathbf{S}_\perp \mathbf{Y} \boldsymbol{\pi} = 0$  for all motions (zero lever arm, zero first moment, zero inertia); massless bodies contribute nothing; with  $\mathbf{u} = 0$  the actuated rows vanish identically. Both the dynamics and the effort terms are then at their global minimum.  $\square$

This degeneracy was reproduced numerically during development (§9): with a weak absolute prior the fit returned  $m_0 = 2.3 \text{ kg}$  (the total) and zeros elsewhere. Two remedies are structural, not cosmetic: the physically consistent manifold (§5), which keeps every iterate realisable, and

anchors supplying inhomogeneous information (known club, total mass, ground wrench). A third is honest reporting: every fit publishes which directions of  $\boldsymbol{\pi}$  came from the data.

**Remark 1** (Effort priors bias inertia downward). *Any penalty on  $\|\mathbf{u}\|$  or  $\|\Delta\mathbf{u}\|$  is informative about  $\boldsymbol{\pi}$  because  $\mathbf{u} = \mathbf{B}^+(\mathbf{Y}\boldsymbol{\pi} - \mathbf{J}_c^\top \boldsymbol{\lambda})$  on the actuated rows. The bias is small when the unactuated rows and anchors dominate the information (the planar result shows  $\leq 3\%$ ) but it must be reported, and model-free kinematic regularisation (a jerk prior on  $\mathbf{q}$ ) is preferable to strong effort priors when acceleration noise is the problem.*

## 5 Physically Consistent Inertia

Physical consistency is the convex cone of parameters whose  $4 \times 4$  pseudo-inertia  $\mathcal{J}(\boldsymbol{\pi}) = \begin{bmatrix} \frac{1}{2} \text{tr}(\mathbf{I})\mathbf{1} - \mathbf{I} & \mathbf{h} \\ \mathbf{h}^\top & m \end{bmatrix} \succeq 0$  (Wensing–Kim–Slotine). Two representations are used: (i) the linear 10-vector with Euclidean projection onto the cone in the pseudo-inertia Frobenius metric (eigenvalue clipping; idempotent; identity on consistent bodies), used to sanitise priors and report violations; and (ii) the unconstrained log-Cholesky coordinates  $\boldsymbol{\theta}$  with  $\mathcal{J} = \mathbf{L}\mathbf{L}^\top$ ,  $\text{diag } \mathbf{L} = e^\theta$  (Rucker–Wensing), used as outer variables so every iterate is realisable. For planar bodies ( $m, h_x, h_y, I_o$ ) the cone is the rotated second-order cone  $mI_o \geq |\mathbf{h}|^2$ , with coordinates  $(\log m, \mathbf{h}, \log I_c)$ ,  $I_c = I_o - |\mathbf{h}|^2/m$ , and a closed-form projection. Because  $\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}$  is available analytically, the dynamics Jacobian in the inertial direction is exact:  $\mathbf{W}\mathbf{Y} \partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}$ .

## 6 Estimation Problem

With trajectory basis  $\mathbf{q}_t^k = \mathbf{P}_t^k \mathbf{C}^k$ ,  $\mathbf{v}_t^k = \mathbf{V}_t^k \mathbf{C}^k$ ,  $\mathbf{a}_t^k = \mathbf{A}_t^k \mathbf{C}^k$  (clamped quintic B-splines; position, velocity and acceleration are all linear in  $\mathbf{C}^k$ ), the reduced objective is

$$\begin{aligned} F(\boldsymbol{\xi}) = & \sum_{k,t} \rho \left( \left\| \mathbf{R}_{kt}^{-1/2} [\mathbf{m}(\mathbf{q}_t^k; s) - \mathbf{y}_t^k] \right\|^2 \right) + \min_w \left[ \sum_{k,t} \left\| \mathbf{W}_\tau [\mathbf{Y}(\mathbf{q}_t^k, \mathbf{v}_t^k, \mathbf{a}_t^k; s) \boldsymbol{\pi}(\boldsymbol{\theta}) - \mathbf{B}\mathbf{u}_t^k - \mathbf{J}_c^\top \boldsymbol{\lambda}_t^k] \right\|^2 \right. \\ & + w_u^2 \sum \left\| \mathbf{u}_t^k \right\|^2 + w_{\Delta u}^2 \sum \left\| \Delta^2 \mathbf{u}_t^k \right\|^2 + w_T^2 \sum \left\| \mathbf{u}_t^k - \bar{\mathbf{u}}_{\phi(k,t)} \right\|^2 + w_\lambda^2 \sum \left\| \boldsymbol{\lambda}_t^k \right\|^2 \left. \right] \\ & + \left\| \mathbf{W}_s(s - s_0) \right\|^2 + \left\| \mathbf{W}_\pi(\boldsymbol{\pi}(\boldsymbol{\theta}) - \boldsymbol{\pi}_0) \right\|^2 + \sum_j w_j^2 (\mathbf{a}_j^\top \boldsymbol{\pi}(\boldsymbol{\theta}) - b_j)^2, \end{aligned} \tag{7}$$

where  $\rho$  is a robust kernel (Huber/Cauchy by IRLS),  $\mathbf{W}_\tau = \text{diag}(\gamma/\sigma_\tau)$  carries the continuation factor  $\gamma \in [0, 1]$  and the declared dynamics discrepancy  $\sigma_\tau$  (which must include the basis discretisation error, §8),  $\mathbf{a}_j^\top \boldsymbol{\pi} = b_j$  are linear anchors (total mass, known club), and  $\bar{\mathbf{u}}$  is the shared phase-indexed template. Everything inside  $\min_w$  is linear in  $w$  for fixed  $\boldsymbol{\xi}$ : the inner problem is a sparse linear least-squares problem (a convex QP/SOCP when input bounds and friction cones are active). This is the *inverse-dynamics collocation* formulation (Remy–Thelen; Falisse *et al.*): no forward integration occurs inside the optimisation, so there is no exponential sensitivity, every node is independent (embarrassingly parallel), and the measured states are feasible initial iterates.

## 7 Algorithm

### 7.1 Variable projection: the “locally learned inputs”

For fixed  $\boldsymbol{\xi}$  the inner minimiser  $w^*(\boldsymbol{\xi})$  is the solution of sparse normal equations  $(\mathbf{A}^\top \mathbf{A})w = \mathbf{A}^\top b$  (block-arrow structure: per-node  $\mathbf{u}$  blocks, a banded  $\Delta^2$  coupling, a small template block). The

inputs are therefore *determined linearly from the kinematics*: this is the exact answer to “is the locally learned set of inputs a linearisation?”—the inputs are a linear function of the inner problem data, not an approximation; the only linearisation in MOSAIC is the Gauss–Newton model of the outer step. Eliminating  $w$  exactly gives the Golub–Pereyra/Kaufman reduced Gauss–Newton system

$$\mathbf{H}_{\text{red}} = \mathbf{J}_{\xi}^{\top} (\mathbf{I} - \mathbf{A}(\mathbf{A}^{\top} \mathbf{A})^{-1} \mathbf{A}^{\top}) \mathbf{J}_{\xi}, \quad \mathbf{g}_{\text{red}} = \mathbf{J}_{\xi}^{\top} r^{\star}, \quad (8)$$

which is the Schur complement of the full Gauss–Newton matrix onto the outer variables (bundle-adjustment structure). The implementation verifies (8) against the explicit projector.

## 7.2 Outer step

Levenberg–Marquardt on  $F$  with Marquardt scaling and Nielsen damping updates (gain ratio  $\varrho$ ;  $\mu \leftarrow \mu \max\{\frac{1}{3}, 1 - (2\varrho - 1)^3\}$  on success, geometric growth on failure). Jacobians: observation rows analytically through  $\partial \mathbf{m} / \partial \mathbf{q} \cdot \mathbf{P}$  and  $\partial \mathbf{m} / \partial s$ ; dynamics rows exactly in  $\boldsymbol{\theta}$  ( $\mathbf{WY} \partial \boldsymbol{\pi} / \partial \boldsymbol{\theta}$ ) and by *vectorised* central differences in  $(\mathbf{q}, \mathbf{v}, \mathbf{a}, s)$ —each perturbation applied to all nodes of all trials at once, so one iteration costs  $2(3n_v + n_s)$  batched regressor evaluations plus one sparse factorisation. Analytic derivatives (Singh–Russell–Wensing; GRiD) replace the differences without changing the algorithm.

## 7.3 Continuation

A schedule  $\gamma \in \{0, 10^{-2}, 10^{-1}, 1\}$  moves the problem from a pure kinematic fit (the current IK practice, now a special case) to the fully weighted dynamics-consistent fit. Each stage is solved to tolerance before the next; the measured receipts (§9) show 8–14 iterations per stage. The homotopy is what makes convergence from an IK seed reliable: the  $\gamma = 0$  basin is convex-like, and the dynamics terms are switched on only once the kinematic residual is at its noise floor.

## 7.4 Convergence properties

For zero-residual problems (synthetic truth, consistent model) Gauss–Newton converges quadratically; with noise the rate is linear with factor governed by residual-weighted curvature over the smallest reduced-Hessian eigenvalue. Weakly identifiable inertial directions make that eigenvalue small; the Levenberg–Marquardt damping with the Yamashita–Fukushima rule  $\mu \propto \|r\|$  retains local quadratic convergence under a local error bound even when the Jacobian is rank-deficient, which is the generic case here. Global convergence to a stationary point follows from the trust-region acceptance test (Moré). Variable projection removes the inner variables’ nonlinearity from the outer iteration and is known to reduce the number of local minima and iterations in separable problems (Golub–Pereyra 2003).

## 7.5 Local policy and replay acceptance

After the fit, the discrete step of the (any-engine) model is linearised along  $(\mathbf{x}^{\star}, \mathbf{u}^{\star})$  by vectorised central differences, and a backward Riccati pass gives time-varying gains  $\mathbf{K}_t$ . The pair  $(\mathbf{u}_t^{\star}, \mathbf{K}_t)$ ,  $\mathbf{u} = \mathbf{u}_t^{\star} - \mathbf{K}_t(\mathbf{x} - \mathbf{x}_t^{\star})$ , is the locally optimal linear policy around the fit—the object a whole-body controller consumes. Acceptance is two-sided: the *open-loop* replay from one initial state is the physical gate (a dynamically consistent fit must replay), and the closed-loop replay reports the feedback effort needed to hold the motion, which quantifies the fit’s sensitivity. For repeated trials the iterative-learning view  $\mathbf{u}_{k+1} = \mathbf{u}_k + \mathbf{L}e_k$  converges iff  $\rho(\mathbf{I} - \mathbf{GL}) < 1$ ; with  $\mathbf{L}$  the Riccati gain this is a cross-trial diagnostic, not the estimator.

## 7.6 Cross-trial torque template

Each trial’s inputs are binned by normalised phase; the trial-balanced mean  $\bar{\mathbf{u}}(\phi)$  enters the inner solve as a coupling prior ( $w_T$ ) and is reported with its principal modes (SVD of trial deviations) as the player’s motor signature. On a planted rank-one structure the first mode explains  $> 99.9\%$  of the variance (test).

## 7.7 Complexity and vectorisation

Per outer iteration:  $O(N)$  batched regressor evaluations ( $N = \sum_k T_k$  nodes), one sparse LU of the inner normal matrix ( $n_w = Nn_u + Pn_u + n_\lambda$ , banded), one dense reduced system of size  $n_\xi = \sum_k C_k n_v + n_s + 10n_b$ . Trials enter only through the shared blocks, so the per-trial parts are independent and batch over trials; the time axis is parallel because no integration occurs. The reference implementation is pure NumPy; the same kernels accept MJX/JAX batched regressors unchanged.

## 8 Qualification Gates

A fit is reported with the following measured items; a missing item is “unqualified”, never “passed”.

1. **Kinematic whiteness.** At  $\gamma = 0$  the whitened marker residual RMS must be within a declared factor of 1 and uncorrelated in time. Under-resolved bases fail this gate; dynamics weighting must not be switched on before it passes (Proposition 3 is reached otherwise, because a basis that cannot represent the acceleration makes the massless body the cheapest consistent explanation).
2. **Declared discrepancy.**  $\sigma_\tau$  must be at least the basis acceleration error times the mass scale; for the planar fixture a 80-coefficient quintic basis over 200 nodes gives 1% acceleration error.
3. **Physical consistency.** Every reported  $\boldsymbol{\pi}$  is on the cone by construction; violations of the sanitised prior are logged.
4. **Observability.** Rank and per-parameter observable fraction of the unactuated-row regressor; posterior information after eliminating inputs; directions supplied by the prior are named.
5. **Open-loop replay.** Configuration RMS and divergence time from one initial state with the fitted inputs; the accepted tolerance is declared beforehand.
6. **Closed-loop effort.** TVLQR feedback RMS relative to the nominal inputs.
7. **Cross-trial consistency.** Template explained variance and the spread of shared parameters across leave-one-trial-out refits.

## 9 Reference Results on the Planar Fixture

Fixture: pinned planar 3-link chain (pelvis–arm–club analogue), first joint unactuated, true lengths (0.9, 0.6, 0.4) m, two trials of 150 nodes at 200 Hz driven by smooth torques of RMS 0.54 N m, six markers with 0.5 mm noise, initial geometry +5%, inertial prior perturbed by  $\pm 10\text{--}15\%$  (projected onto the cone), anthropometric  $\sigma = 15\%$ , total mass and club inertia anchored,  $\sigma_\tau = 0.2$  N m,  $w_u = 0.3$ ,  $w_{\Delta u} = 10$ . Measured on NumPy 2.5/SciPy 1.18, single core:

|                                                          |                                             |
|----------------------------------------------------------|---------------------------------------------|
| Initialisation (sequential IK, 2 trials)                 | 1.0 s                                       |
| Joint fit (4 stages, 8+9+9+11 iterations, all converged) | 3.1 s                                       |
| Replay + TVLQR (2 trials)                                | 1.8 s                                       |
| Geometry error (identifiable lengths)                    | < 0.1 mm                                    |
| Kinematic RMS error                                      | 0.5 mrad                                    |
| Torque RMS error / torque RMS                            | 0.106/0.542 = 20 %                          |
| Inertial parameters (observable, vs. truth)              | 1–3 % (prior: 4–10 %)                       |
| Structural observability rank                            | 9 of 12 ( $m_0$ unobservable, as predicted) |
| Open-loop replay, configuration RMS over 0.75 s          | 0.030 rad, no divergence                    |
| Closed-loop replay / feedback effort RMS                 | 0.0005 rad / 0.02–0.03 N m                  |

Negative results retained: (a) with a 16- or 32-coefficient basis the fit converged to the massless body (gate 1 failure); (b) with  $\sigma_\tau = 0.05$  N m and no input smoothness the torques were dominated by acceleration noise (RMS error > 1 N m); (c) with a weak absolute prior and no club anchor the scale gauge was resolved by the degenerate solution. Each is now a test or a gate.

### Reproduction.

```
export PYTHONPATH=$PWD:$PWD/src:$PWD/src/shared/python
python3 -m pytest tests/unit/estimation/mosaic -q
```

## 10 Engine Integration Path

- **Regressor provider.** Implement `RegressorModel` for Pinocchio (`computeJointTorqueRegressor`, batched over nodes in C++), MuJoCo (`mj_inverse` differentiated w.r.t. body inertias, or MJX autodiff with `vmap`), Drake (`AutoDiffXd`). Marker forward kinematics and Jacobians come from the same engine.
- **Floating base and manifold.** Replace the Euclidean spline by a spline in local coordinates with retraction (Solà); the inner solve is unchanged because it never differentiates  $\mathbf{q}$ .
- **Contact.** Add  $\boldsymbol{\lambda}$  to  $w$  with friction-cone/unilateral constraints (the inner problem becomes a small SOCP; use an active-set or proximal solver; the reduced system is unchanged on the free set). Treat measured GRF as observations of  $\boldsymbol{\lambda}$ , never as decision variables.
- **Observations.** Markerless 2-D keypoints enter through camera projection with robust kernels; marker attachment offsets are geometry variables so dimension refinement propagates to the markers.
- **Actuation layer.** Net torques first; an optional declared joint-level activation model maps torques to bounded activations with first-order dynamics (reported as a model choice, not a measurement).

## 11 Generalisation to Humanoids and Manipulation

The kernels are model-agnostic: a humanoid’s floating-base rows supply (6), a known payload or the measured ground wrench supplies the anchor, and the output  $(s, \boldsymbol{\pi}, \mathbf{u}^*, \mathbf{K})$  is a reference-plus-gains schedule that whole-body controllers consume directly. For manipulation the object plays the club’s role (unknown inertia identified from the wrench transmitted through a known-inertia tool, or vice versa). Identified parameters become the nominal model; residual learning or light randomisation covers what remains, rather than replacing identification.

## 12 Review of the DIME Epic and Repository Defects

The audit of #11421 (child issues #11422–#11437, all merged 2026-10-02..04) found that the estimator contracts and bookkeeping landed but the estimator largely did not. Recorded for correction, with file references at audit commit 21a2ed06:

- No engine backs any DIME provider; the `DynamicsProvider` protocol lacks derivatives, parameter setters and batching (`dime_contracts.py`:930-956).
- The “coupled state-control window” is control-only single shooting with hard-coded zero defects and costs (`dime_dynamics_window.py`:517-566,604).
- MHE never applies its arrival factor (`moving_horizon.py`:140-213,401,636).
- Learned initialisers, ablations, baseline and replay audit return constants (`dime_learned_initializers.py`:3, `dime_benchmark_ablations.py`:214-255, `dime_manifest.py`:563-629, `dime_continuous_replay.py`:3; solver-cache timings are fabricated and its key omits the targets (`dime_solver_cache.py`:293-313,408-411).
- `multi_trial.py`:379 vs :413-426: locked parameters with a prior give a residual/Jacobian row mismatch; its covariance is conditional, not marginal.
- `drift_prediction.py`: covariance propagation omits  $\partial f/\partial \mathbf{x}$ ; a horizon collapses to one step; the default selection matrix actuates the root.
- `dime_manifold.py`: `local_coordinates_jacobian` is the identity; the retraction Jacobian is valid only at zero velocity.
- ZVCF is defined two incompatible ways (`ztcf_zvcf.py`:339 vs `counterfactual.py`:62,98); `_dynamics_interface.py`:227 has a sign error; the GS3DX reference’s ZTCF equation carried an undefined  $\lambda$  under a “contact free” heading and integrated the drift along the measured state (corrected in that document alongside this one).
- No inertial regressor existed anywhere in the estimation package.

## 13 Roadmap

The follow-up epic (MOSAIC) sequences: engine regressor providers and a conformance test against this reference; floating-base manifold spline; contact wrenches with cones and GRF observations; 2-D keypoint observations and attachment-offset geometry; capture-A/capture-O fits with the gates of §8; multi-swing acquisition (the repository holds one trial per capture); analytic derivatives and MJX batching; activation layer; and remediation of the defects above. Promotion criteria are frozen before the experiments: open-loop replay tolerance, torque RMS against synthetic truth, and time-to-accepted-match against the IK+smoothing baseline, all measured.

## 14 Non-Claims

This document does not claim real-time performance, a global optimum, muscle force identification, or validity of any human fit. Synthetic truth is truth of the simulator. Numbers quoted are measured on the stated fixture and hardware and are reproducible by the stated command.

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